L6790: Validation status and preliminary results

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Preliminary Specs

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LLC-PFC controller IC for resonant insulated PFC preconditioners

Key Features

HB controller implementing HB or FB topology with automatic morphing

800 V Integrated High-Voltage startup

2-level OCP with autorestart and frequency shift

Anti-capacitive mode protection

“Safe-start” procedure preventing hard switching at startup

Bin/out, OVP and OTP protections

Adaptive dead-time control

Optimized burst-mode operation at light load

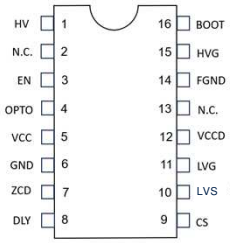
Programmable input voltage brown-in/brownout thresholds for single or universal range applications

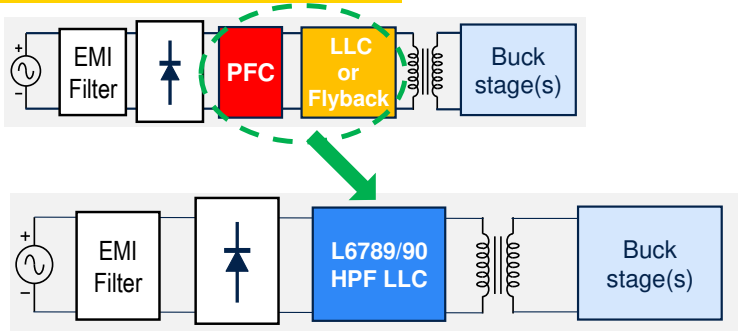
IC disable function

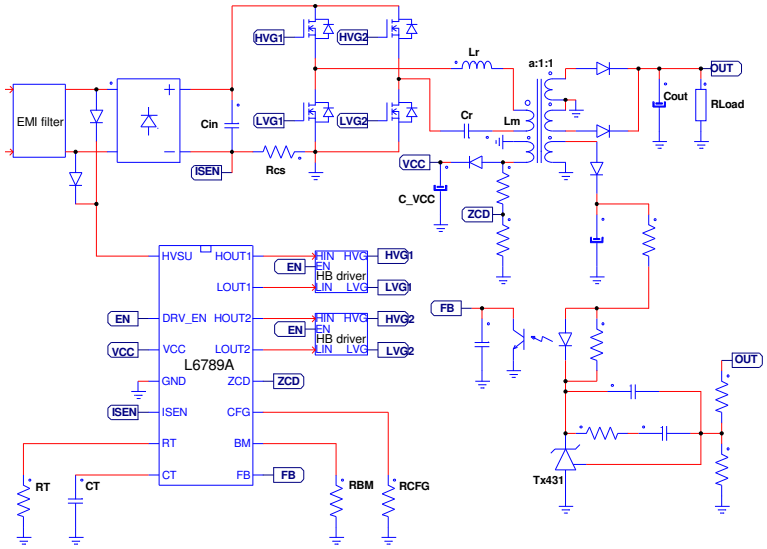
X-capacitor discharge



SO16N

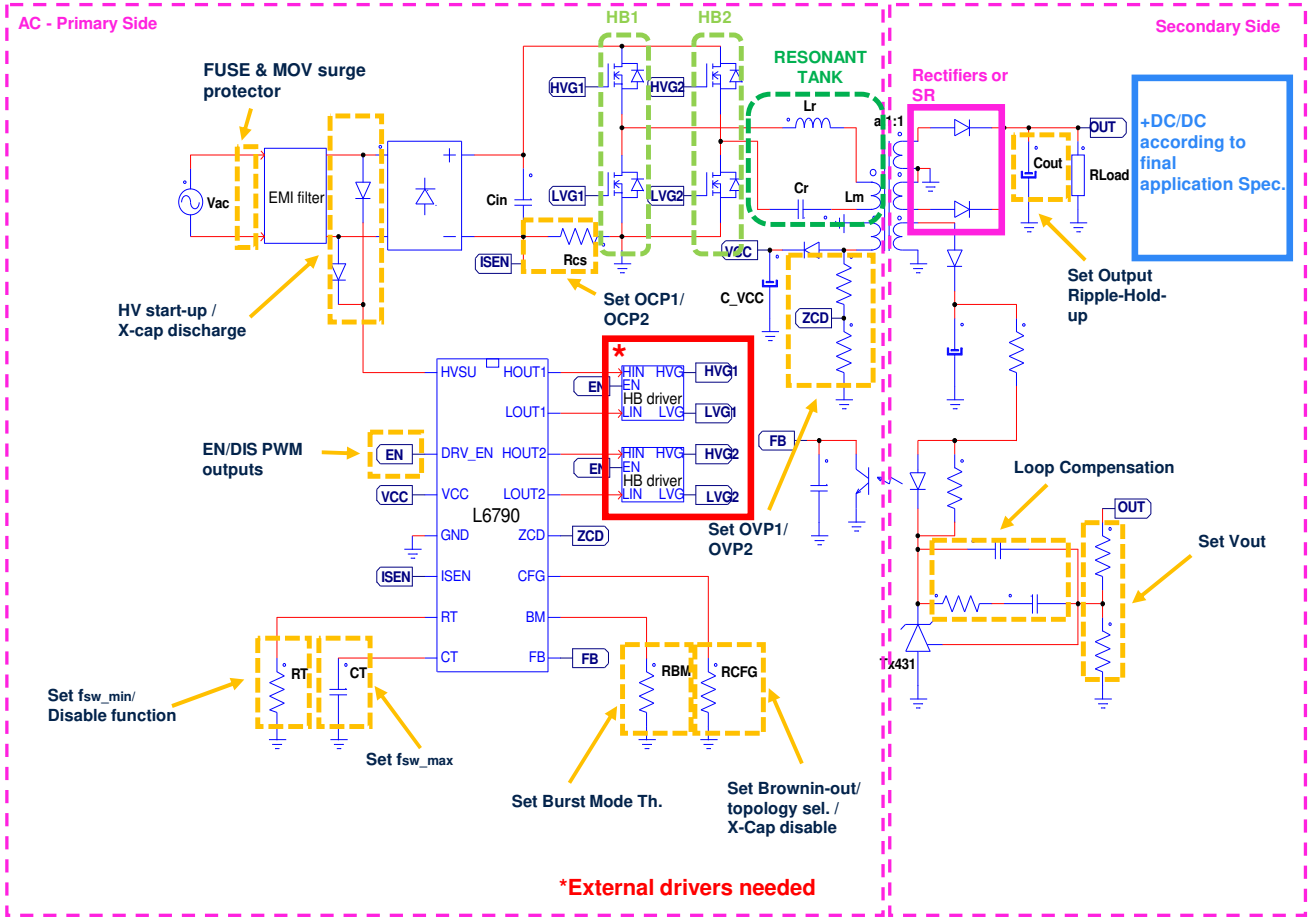






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Typical Application Block Diagram



Supply voltage setting-VCC cap	$C_{VCC_min} = 4\frac{(I_{CC_{drivers}} + I_{CC})}{VCC_{ON} - VCC_{HVSU_ON}}$
Maximum input power setting	$R_{SENSE} = \frac{16.8\Omega W}{Pin_{MAX}}$
Switching frequency window during operation and at start-up	$f_{min} = \frac{1}{2(C_T + R_T + T_{idle})}$ $f_{max} = \frac{1}{2(\frac{C_T}{\frac{I_{EA}}{V_{CT_ref}} + \frac{1}{R_T}} + T_{idle})}$
OVP functions setting	$V_{OUT} = \frac{n_S}{n_A} V_{ZCD} \left(\frac{R_{ZCD_H}}{R_{ZCD_L}} + 1\right)$
Brown-in, Brown-out setting	$BO = R_{CFG} 4 \frac{V}{k\Omega}$ $BI = BO + 10V$
Burst Mode level setting	$R_{BM} = \frac{16.7k\Omega}{V^2} R_{CS} P_{IN_BM}$

➢ **Extremely minimum part count required (orange blocks) makes it easy to be used**

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Timeline and Validation Status



AA silicon version Open Point to be addressed in BA version:

1. HVSU start-up current oscillation during VCC charging→ not affecting the final application
2. P-ON Reset issue→ application could stop unexpectedly (mitigation action analysis on going)

→Application/Design Evaluations are on going
→ AA-ES not available for preliminary customer evaluation due to point 2

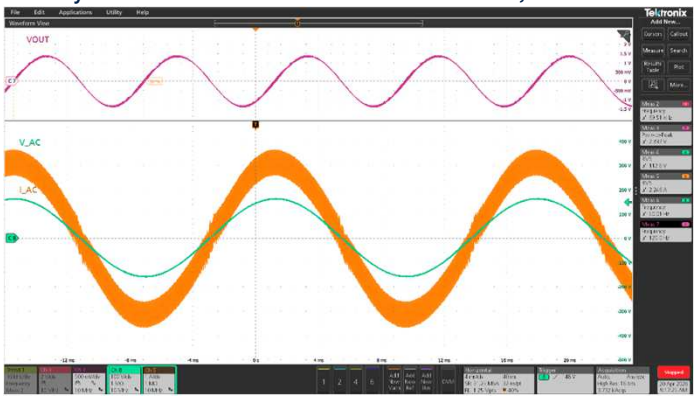
240W-60V Prototype preliminary waveforms

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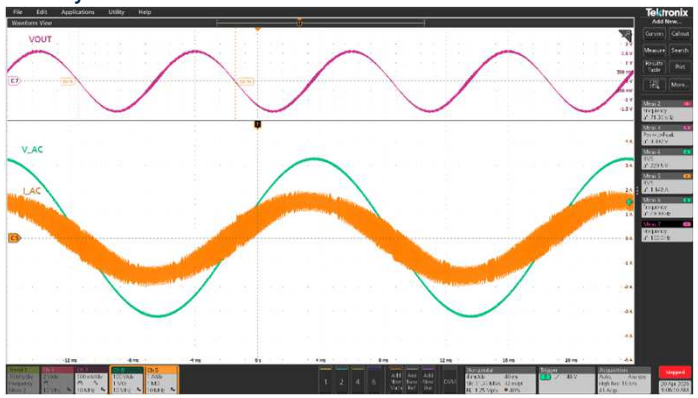
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PFC main waveforms@ nominal mains voltage

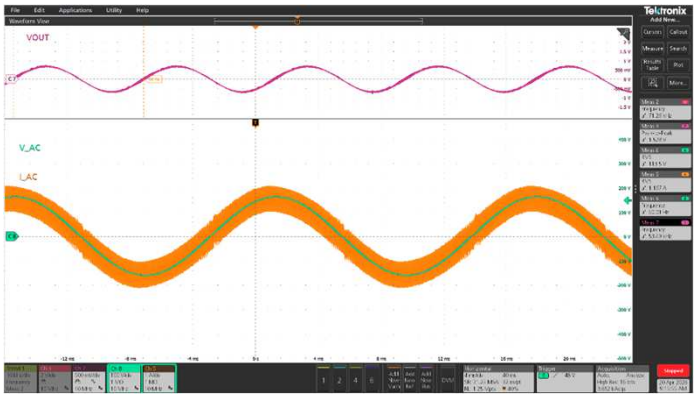
Steady state: **115Vac**/60Hz-60V/4A- **THD=1,75%**



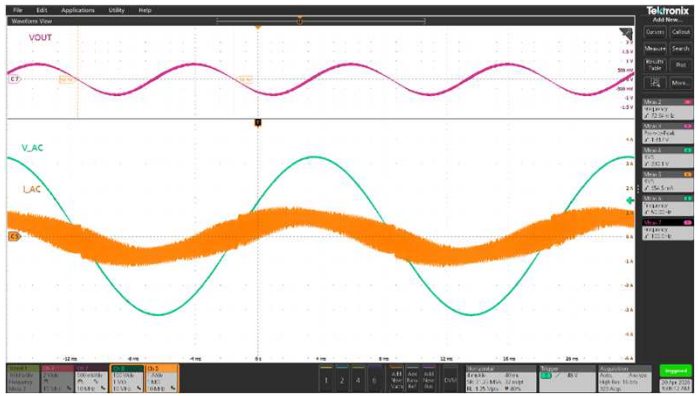
Steady state: **230Vac**/50Hz-60V/4A- **THD=3%**



Steady state: **115Vac**/60Hz-60V/2A-**THD=2,4%**



Steady state: **230Vac**/50Hz-60V/2A-**THD=5,5%**

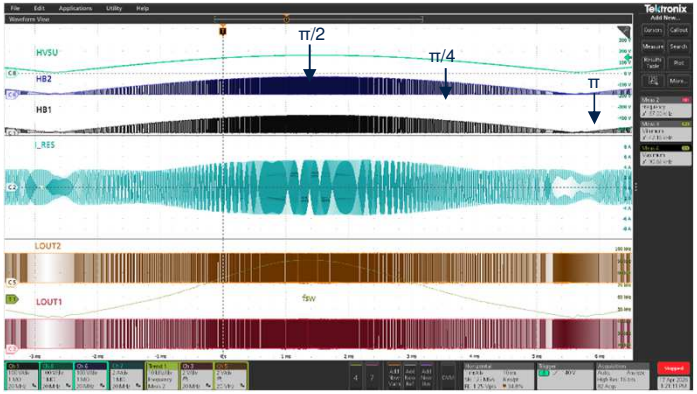


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Resonant Current variation @115Vac/60Hz

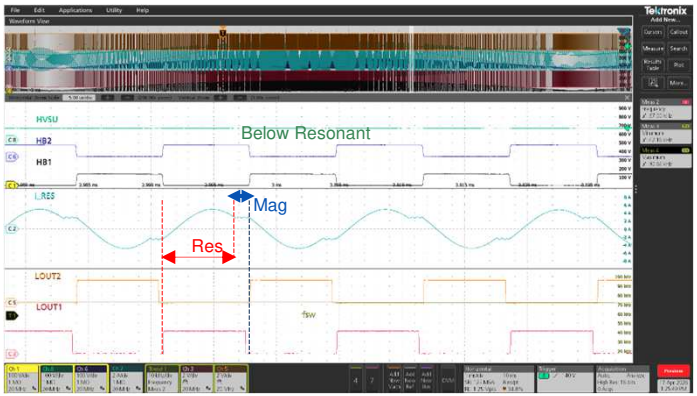
Half Cycle@115Vac-full load- **FULL BRIDGE**



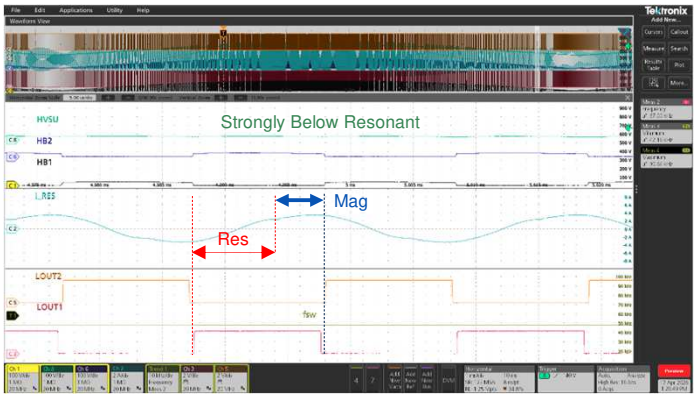
Resonant Current on the peak (π/2) - fsw=93,03kHz



Resonant Current on π/4 - fsw=71,3kHz



Resonant Current close to Zero Crossing (π)- fsw=47,2kHz

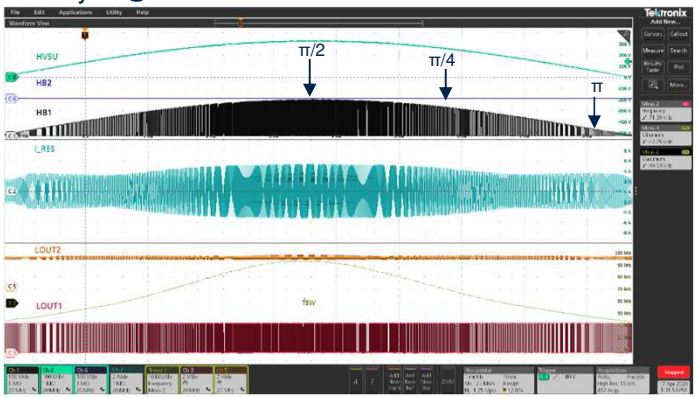


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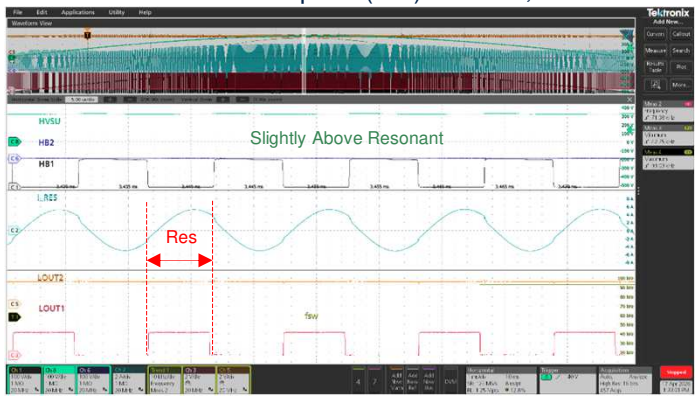
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Resonant Current variation @230Vac/50Hz

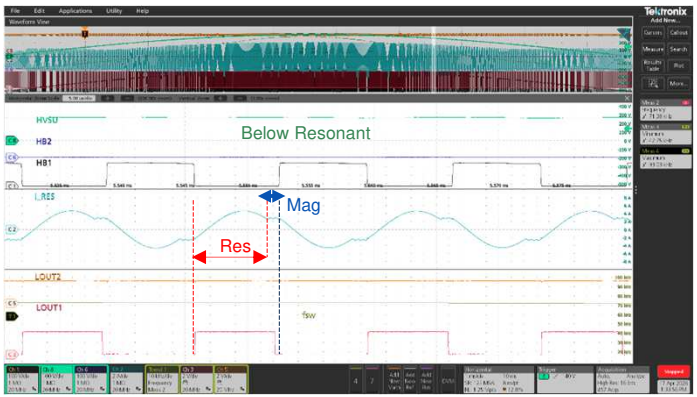
Half Cycle@230Vac-full load - **HALF BRIDGE**



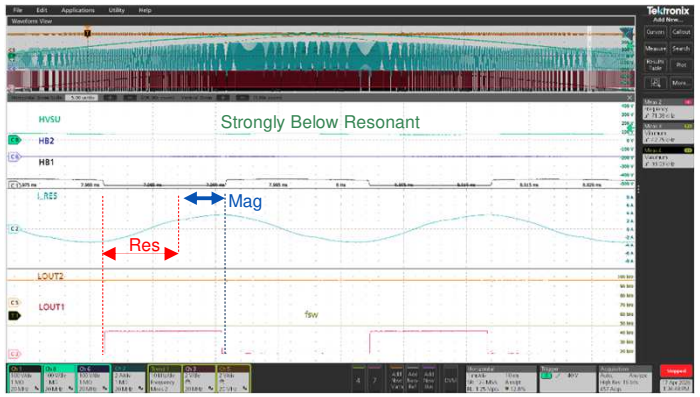
Resonant Current on the peak (π/2) - fsw=93,03kHz



Resonant Current on π/4 - fsw=71,3kHz



Resonant Current close to Zero Crossing (π)- fsw=47,2kHz

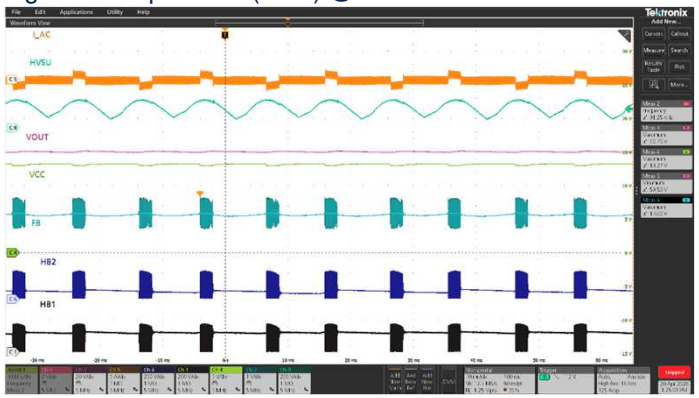


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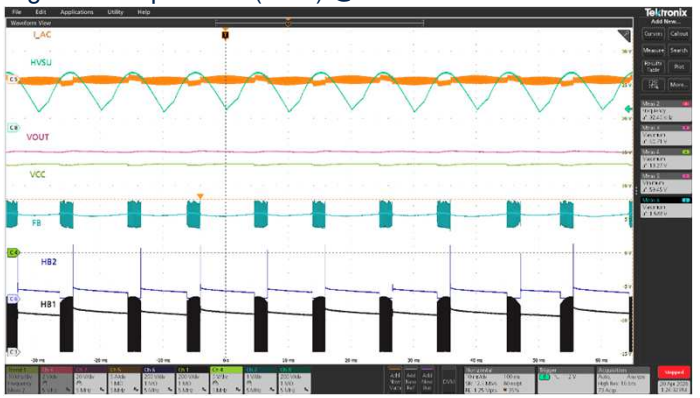
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Light Load-Burst Mode operation

Light Load operation (10%) @115Vac



Light Load operation (10%) @230Vac

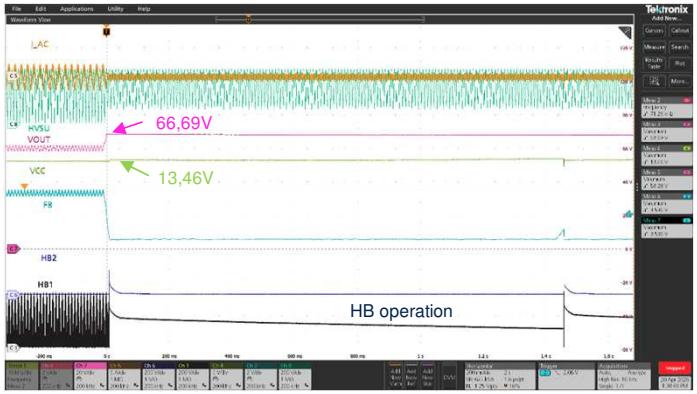


Proprietary Burst Mode feature: synchronized BM packet with the peak of input voltage.

Load variation@115Vac: from 100% to 0% load



Load variation@230Vac: from 100% to 0% load



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Promotional Tools

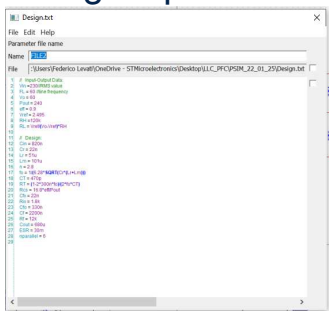
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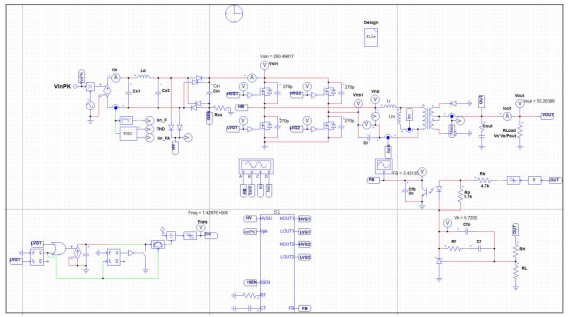
L6790 design tools

✓ PSIM model: available upon request

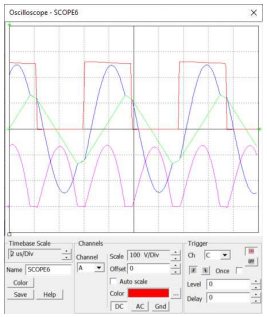
Design Spec.



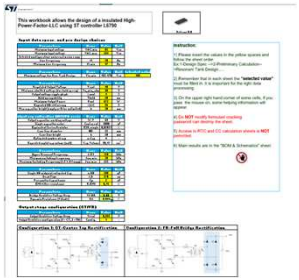
Schematics

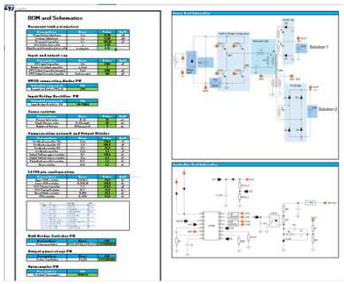


Electrical waveforms



✓ Excel Spreadsheet: preliminary version available upon request



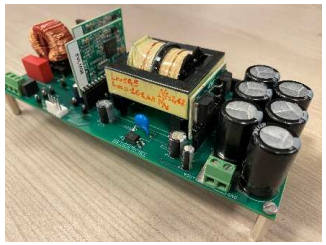


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L6790 evaluation board

60V-240W single stage LLC-PFC

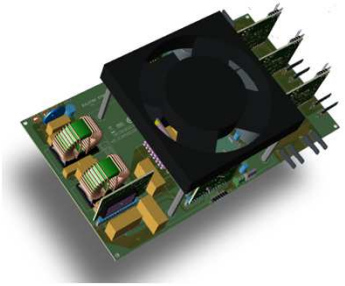


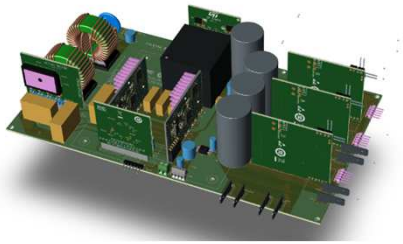
LLC-PFC converter:
- VIN = 90 ~ 265 VAC
- VOUT = 60 V DC
- POUT = 240 W
- THD < 10 %
- Efficiency = 95%@115Vac

Order code:
EVL6790-240W

Available in 2027

48V-670W single stage LLC-PFC+DC-DC





LLC-PFC converter:
- VIN = 90 ~ 265 VAC
- VOUT = 48 V DC
- POUT = 670 W

+
BUCK1:24V-21A
BUCK2:20V-3A
BUCKK3:12V-8A

Order code:
EVL6790-670W

Available in 2027

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Thank you

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